Geometry and Topology for Modern Applications

Smooth manifolds, algebraic topology, topological data analysis, geometric deep learning

Bikash Thakur

May 29, 2026

Abstract

We will cover geometry and topology from manifolds and cohomology to TDA, information geometry, and geometric deep learning.

Contents

1 PART I — SMOOTH MANIFOLDS AND RIEMANNIAN GEOMETRY

1.1 1. Smooth Manifolds

- Charts, atlases, smooth maps; transition functions
- Tangent spaces and tangent bundle; differential of a map
- Vector fields; flows; Lie bracket
- Submanifolds; Whitney embedding theorem; examples (spheres, Lie groups, Grassmannians)

1.2 2. Differential Forms and Stokes' Theorem

- Exterior algebra; k-forms; wedge product
- Exterior derivative; Poincaré lemma; de Rham cohomology
- Stokes' theorem (general); Green, Gauss, classical Stokes as special cases
- Integration over chains; orientation

1.3 3. Vector Bundles and Connections

- Vector bundles; trivial and non-trivial; sections
- Connections; parallel transport; holonomy; geodesics
- Curvature as obstruction to parallel transport; curvature 2-form

1.4 4. Riemannian Geometry

- Riemannian metrics; Levi-Civita connection; geodesic equation
- Exponential map; normal coordinates; cut locus
- Curvature: sectional, Ricci, scalar; geometric meaning
- Comparison theorems: Bonnet-Myers; Rauch; applications

1.5 5. Lie Groups and Symmetric Spaces

- Lie groups: $SO(n)$, $SU(n)$, $SE(n)$, $Sp(n)$, $GL(n)$
- Lie algebra; exponential map; adjoint representation; BCH formula
- Homogeneous spaces; Riemannian symmetric spaces
- Applications: statistics on SPD matrices, Grassmannians, robotics

1.6 6. Information Geometry

- Statistical models as Riemannian manifolds; Fisher information metric
- α-connections; duality; Amari's framework
- Dually flat manifolds; exponential and mixture families
- Natural gradient descent; connections to optimization (Course 5)
- Cramér-Rao bound from the geometric perspective

2 PART II — ALGEBRAIC TOPOLOGY

2.1 7. Homotopy Theory

- Homotopy equivalence; fundamental group; van Kampen theorem
- Covering spaces; deck transformations; universal cover
- Higher homotopy groups (brief); Whitehead theorem
- Applications: robotics; configuration spaces; topological phases of matter

2.2 8. Homology and Cohomology

- Simplicial complexes; chain complexes; simplicial homology
- Singular homology; Mayer-Vietoris; long exact sequence
- De Rham cohomology; de Rham theorem
- Poincaré duality; applications to manifold topology

2.3 9. Characteristic Classes

- Vector bundles; classifying spaces; universal bundles
- Chern classes (complex); Pontryagin classes (real); Whitney sum formula
- Euler class; Gauss-Bonnet-Chern theorem
- Applications: topological insulators; anomaly cancellation; index theorem (brief)

3 PART III — TOPOLOGICAL DATA ANALYSIS

3.1 10. Persistent Homology

- Filtrations of simplicial complexes; nested inclusions
- Persistence modules; barcodes and persistence diagrams
- Stability theorem: bottleneck distance; 1-Wasserstein stability
- Algorithms: Vietoris-Rips; Čech; alpha complexes; Ripser, GUDHI

3.2 11. Topological Summaries

- Persistence images; persistence landscapes; Betti curves
- Mapper algorithm: topological summaries of high-dimensional data
- Euler characteristic transforms; integral geometry
- Applications: neuroscience; materials science; biology; time series

3.3 12. Sheaves on Networks

- Sheaves on simplicial complexes; sheaf cohomology
- Sheaf Laplacian; cellular sheaves
- Applications: sensor networks; distributed inference; GNNs

3.4 13. Topology and Machine Learning

- Topological loss functions for neural networks
- Topology of loss landscapes; critical points
- Manifold hypothesis; intrinsic dimensionality estimation
- Persistent homology for time series and natural language

4 PART IV — GEOMETRIC MECHANICS AND DEEP LEARNING

4.1 14. Symplectic Geometry

- Symplectic manifolds; Darboux theorem
- Hamiltonian vector fields; Poisson brackets; Liouville's theorem
- Moment maps; symplectic reduction; Marsden-Weinstein
- Symplectic integrators (connection to Courses 3 and 4)

4.2 15. Discrete Differential Geometry

- Discrete exterior calculus on simplicial meshes
- Discrete curvature: angle defect; mean curvature via Laplace-Beltrami
- Cotangent Laplacian; applications to geometry processing
- Surface PDEs and heat flow on meshes

4.3 16. Geometric Deep Learning

- Symmetries as geometric priors; equivariance as a design principle
- Group actions on function spaces; invariant and equivariant maps
- Graph neural networks: message passing as discrete exterior calculus
- Gauge-equivariant networks; principal bundle formalism
- $SO(3)$ and $SE(3)$ equivariant architectures: spherical CNNs, E3NN, NequIP
- Connections to representation theory (Course 2, Sections 32–33)